Applications Index

Business and Economics

Cobb-Douglas production function, 833
Cost, 146
Economic growth, 130
Inventory management, 141, 243
Inventory replenishment, 243
Marginal cost, 146–147, 231, 306
Marginal profit, 236–238
Marginal rates and taxes, 147
Marginal revenue, 147, 236–238
Maximizing utility, 833–834
Maximum profit, 236
Order size, 815
Production growth, 197–198

Engineering and Physical Sciences

Acceleration, 144, 154, 191, 258
Aircraft flight distance, 175, 177
Airplane ground speed and direction, 678
Airplane takeoff, 151
Angle for pipes, 463–464
Architecture, 306
Area, 872
Atmospheric pressure and altitude, 423
Average temperature, 165, 867
Beam motion, 180, 196
Beam under a load, 889
Bouncing ball, 546
Boyle's Law, 31
Catalyst for chemical reaction, 243
Center of mass, 886
  of thin shell, 982–983
Centroid, 886, 897–898
Chemical mixture problem, 1033–1035
Chemical reaction, 423, 497
Circulation
  around bounding circle, 988–989
  density, 955–956
  of paddle wheel, 992
Converting mass to energy, 189
Cooling, 422
Cylinder pressure, 141
Damped motion, 17-17–17-18
Displacement
  vs. distance traveled, 271–272
  of an object, 142–143
Distance traveled, 269–271, 274–275
Draining a storage tank, 151, 196
Draining a swamp, 510
Electrical resistance, 815
Electric circuit, 17-18–17-19, 1028–1029, 1043
Electricity, 177
Electromagnetic theory, 1015
Elevation angle of a hot air balloon, 173, 178
Escape velocity, 1050
Falling object, 149, 165, 252–253
  resistance, 1040–1041
  skydiving, 1042
Filling a conical tank with water, 172
Flight navigation, 436
Fluid force
  on a curved surface, 1015
  out of hole in water tank, 263
  streamlines, 965
Flux, 938, 961, 980, 981
Free-falling object, 56–58, 144–145, 168, 214, 258–259
Glider position, 726–727, 735
Gravitational fields, 944
Hanging cables, 453
Harmonic motion, 17-16–17-17
Heat capacity of a gas, 525
Heat diffusion equation, 1010
Heat equation, 788, 849
Heating a plate, 177
Heat transfer (cooling), 422
Height
  of a baseball, 730, 733
  of a rocket, 1051
Impedance in a series circuit, 195
Inertia, moments of, 887–889
Inflating balloon, 151, 178
Kinetic and potential energy, 31
Length
  of an astroid, 355
  of curves, 356
Mass, 885–886
Maximum volume
  of a box, 232–233
  of a can, 233–234, 812
Measuring cup stripes, 117
Motion
  of an object along a line, 231
  of a helicopter, 697–698
  of a particle, 165, 179, 196, 265
  in the plane, 179
  of a projectile, 148–149, 198, 275, 302, 733
  of a vehicle, 179, 674
Moving shadow, 179
Orthogonal trajectory of curves, 1032–1033
Parametrization
  of a cone, 966
  of a cylinder, 966
  of a sphere, 966
Particle motion, 165, 174–175, 179, 196
Path
  of a ball, 729–730, 733
  of a baseball player, 179–180
  of a football, 263–264
  of a glider, 726–727, 735
  of a javelin, 755
  of a projectile, 728–729, 732, 733
  of a shot-put throw, 732
Planetary orbits, 661
Pressure of fluid
  at bottom of dam, 365
  against vertical flat plate, 366–367
Pressure waves produced by tuning fork, 22
Projectile motion, 148–149, 198, 733
Pumping liquids from containers, 364–365
Radioactive decay, 421, 424
Reservoir water, 260
Resistors connected in parallel, 195, 781
Resultant force, 678–679
Satellite orbit, 369
Skaters stopping distances, 1031, 1035–1036
Skylab 4 view, 756
Sliding ladder, 177
Sonobuoy submarine location, 249
Specific heat of a gas, 327
Speed
  of light, 235–236
  of particle, 196
  stopping distance, 151, 258
  of vehicle, 150, 174, 179
  velocity and, 142–143
Stream flow, 802–803
Submarine intercept, 712
Surface area
  of a cone, 968
  cut from bottom of paraboloid, 971–972
  of a infinite paint can or Gabriel's horn, 522
  of an oil spill, 180
  of a sphere, 969
Suspension bridge, 654

Temperature
air, 327, 1039
below Earth's surface, 764
greatest, 931
mean, 525
and period of a pendulum, 165
Thermal expansion in laboratory equipment, 117
Torque, 692–693, 712
Velocity
and acceleration, 144–146
of weight on spring, 154
fields, fluid, 936–937, 958
in a resisting medium, 1030–1031
of a rocket, 149–150
of a skydiver, 453
of two particles, 198
Vibrating spring, 17-15–17-16
Volcanic lava fountains, 151
Voltage changes, 177, 797
Volume
of a balloon, 37
of fluid in a storage tank, 812
of an ice cream cone, 900–901
of a solid region, 871
of a swamp, 510
of a tetrahedron, 879–880
of a torus (doughnut), 974
of a wedgelike solid beneath surface, 861
Water flow, 116–117
Wave equations, 787–788
Weight on a pulley, 176
Work
along different paths, 952
done by a force field, 935–936
done by a jack, 362, 385
done by a locomotive, 689
done by the conservative field, 946
done in lifting a bucket, 364
required to compress a spring, 363–364, 385

General

Box dimensions, 797, 822
Computer graphics
hidden lines, 704
perspective, 703–704
Endangered species, 424
Leibniz's Rule, 848
Mixtures, 1033–1036
Period of a clock pendulum, 199
Random number generator, 890–891
Sugar, inversion of, 423
Swing of a clock pendulum, 620–621
U.S. Postal Service box size, 239–240
Well depth, 188
Working underwater, 424

Life Sciences

Antihistamine effects, 512
Blood flow, 190–191
Blood types, 834
Body surface area, 151
Cardiac output, 179, 846
Cholera bacteria, 424
Diseases, dying out of, 419–420
Drug assimilation, 512
Drug concentrations, 191, 424
Drug dosages, 615
Endangered species, 424
Flow velocity of a cough, 243–244
Human evolution, 423
Medicine dosages, body's reaction to, 141
Population
density of bacteria, 867
growth, 424
of bacteria, 151, 424
of fruit flies, 60–61, 130–131
of rabbits and foxes, 194–195, 1048–1049
of trout *vs.* bass, 1044–1047
of world, 1031–1032
of yeast culture, 418–419
Transport through cell membrane, 1051

Social and Behavioral Sciences

Information diffusion, 497, 1042
Population, 1041–1042
World population, 1031–1032